

# KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based

[arxiv.org/abs/2504.03553](https://arxiv.org/abs/2504.03553)

Tagline: enable LLM agents to autonomously regulate when to act, reflect, or retrieve knowledge.

Core idea: learn situational self-awareness with lightweight special tokens.

# Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection

Existing agent learning resembles unconscious pattern-fitting.

Three paradigms often apply a 'flood irrigation' strategy:

- Trajectory imitation (e.g., ReAct): inject gold trajectories regardless of need.
- Trial-and-error refinement (e.g., Reflexion): inject external feedback regardless of need.
- Knowledge-augmented planning (e.g., KnowAgent, WKM): inject domain knowledge broadly.

Consequences: fragile planning, pattern collapse on unexpected inputs, and high inference cost.

## Human metacognition

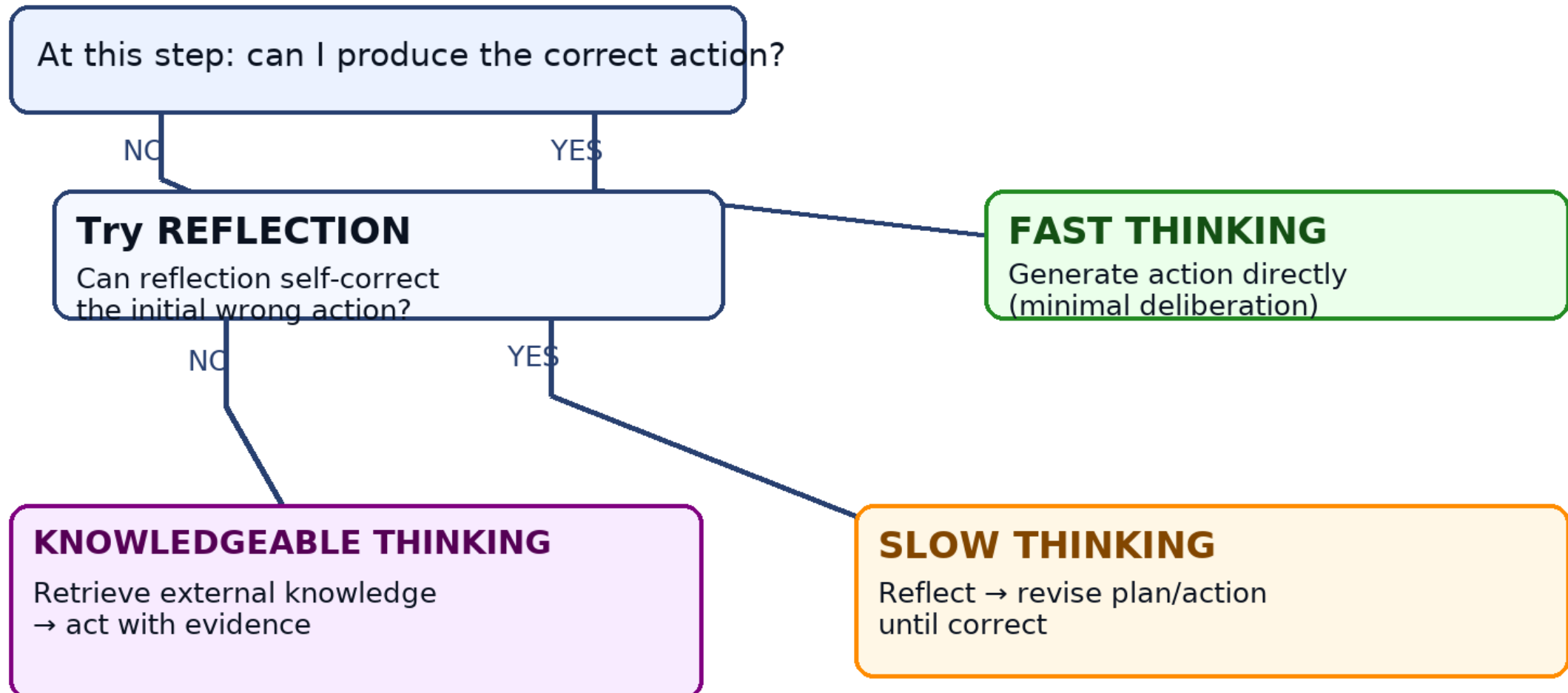
Assess situation → choose cognitive mode:

- act directly
- pause & reflect
- seek external help

KnowSelf operationalizes this as special-token self-awareness.

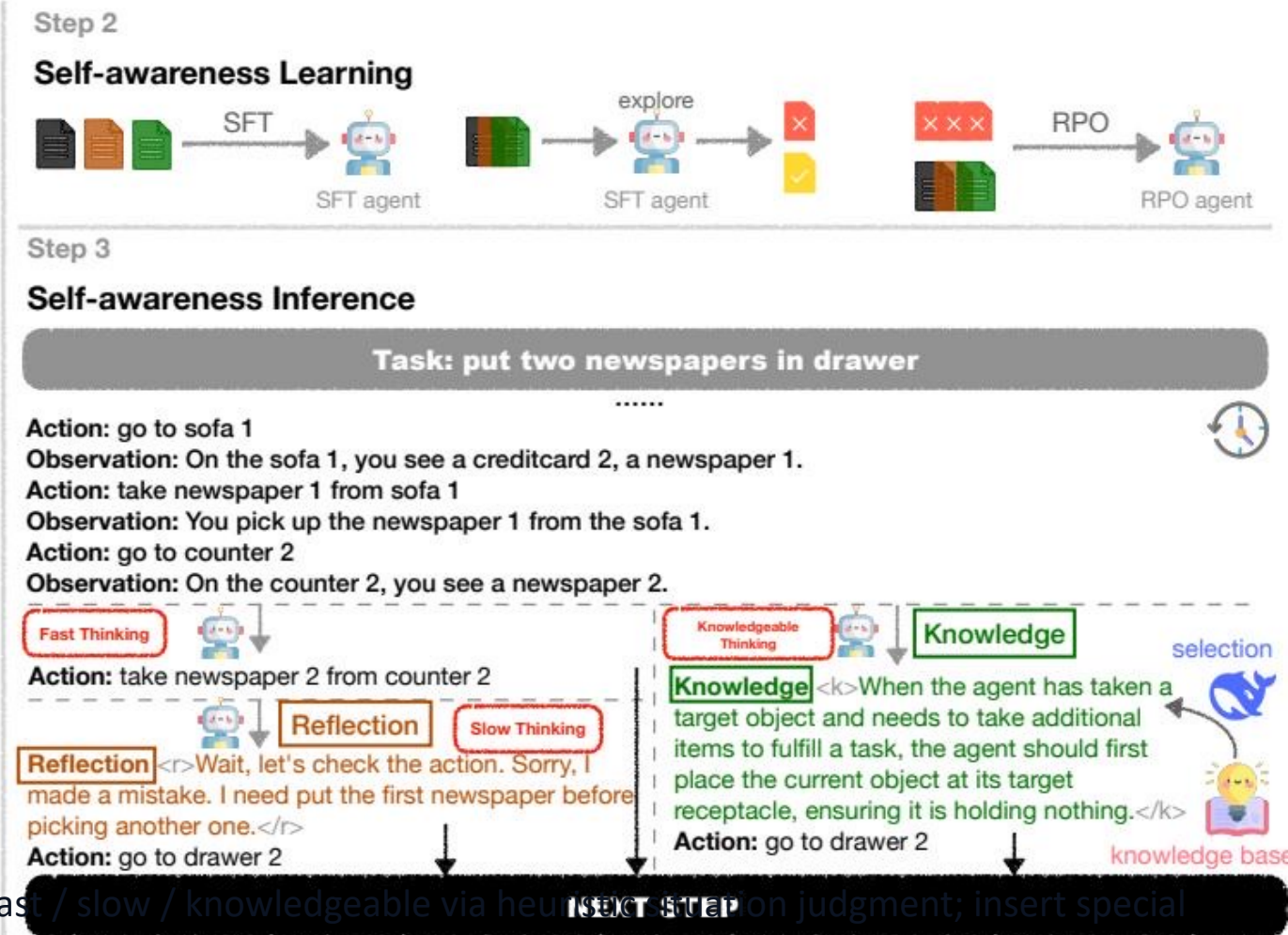
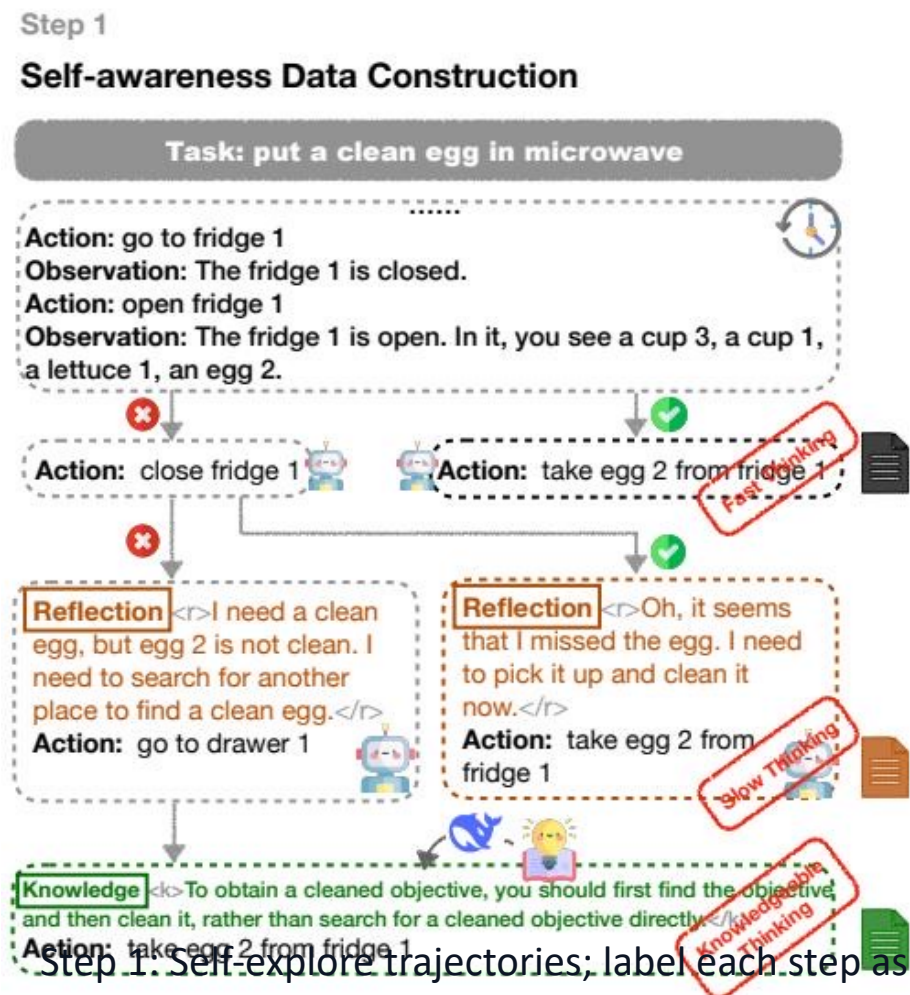
# Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable

## Three Situational Thinking Modes



Heuristic criterion used during data construction: match gold → fast; reflection fixes → slow; else → knowledgeable.

# KnowSelf Framework: Data Construction → Two-Stage Training →



- Step 1: Self-explore trajectories; label each step as fast / slow / knowledgeable via heuristic judgment; insert special tokens.
- Step 2: Two-stage learning: SFT for patterns, then RPO to refine token decisions.
- Step 3: Self-aware inference: generated tokens trigger direct action, reflection, or external knowledge retrieval.



# KnowSelf Correctly Heats a Mug and Places a Saltshaker Where B

| (a) Task: put a hot mug in cabinet   |                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                                                                                                                                                                                                                                                        | OpenAI-O1   | KnowSelf (Llama-8B)                                                                                                                                                                                                     |
| .....                                |                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                         |
| Action:                              | take mug 1 from countertop 1                                                                                                                                                                                                                           |             | Action: take mug 1 from countertop 1                                                                                                                                                                                    |
| Observation:                         | You pick up the mug 1 from the countertop 1.                                                                                                                                                                                                           |             | Observation: You pick up the mug 1 from the countertop 1.                                                                                                                                                               |
| Action:                              | go to microwave 1                                                                                                                                                                                                                                      |             | Action: go to microwave 1                                                                                                                                                                                               |
| Observation:                         | The microwave 1 is closed.                                                                                                                                                                                                                             |             | Observation: The microwave 1 is closed.                                                                                                                                                                                 |
| Action:                              | open microwave 1                                                                                                                                                                                                                                       |             | Action: open microwave 1                                                                                                                                                                                                |
| Observation:                         | You open the microwave 1. In it, you see an egg 1.                                                                                                                                                                                                     |             | Observation: You open the microwave 1. In it, you see an egg 1.                                                                                                                                                         |
| [Knowledge]                          | <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> |             |                                                                                                                                                                                                                         |
| Action:                              | take egg 1 from microwave 1                                                                                                                                                                                                                            |             | [Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r>                                                                             |
| Observation:                         | Nothing happens.                                                                                                                                                                                                                                       |             | Action: heat mug 1 with microwave 1                                                                                                                                                                                     |
|                                      |                                                                                                                                                                                                                                                        |             | .....                                                                                                                                                                                                                   |
| (b) Task: put a saltshaker in drawer |                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                         |
|                                      |                                                                                                                                                                                                                                                        | DeepSeek-R1 | KnowSelf (Llama-8B)                                                                                                                                                                                                     |
| .....                                |                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                         |
| Action:                              | take saltshaker 1 from countertop 3                                                                                                                                                                                                                    |             | Action: take saltshaker 1 from countertop 3                                                                                                                                                                             |
| Observation:                         | You pick up the saltshaker 1 from the countertop 3.                                                                                                                                                                                                    |             | Observation: You pick up the saltshaker 1 from the countertop 3.                                                                                                                                                        |
| Action:                              | go to drawer 1                                                                                                                                                                                                                                         |             | Action: go to drawer 1                                                                                                                                                                                                  |
| Observation:                         | The drawer 1 is open. In it, you see a fork 1.                                                                                                                                                                                                         |             | Observation: The drawer 1 is closed.                                                                                                                                                                                    |
| Action:                              | put saltshaker 1 in/on drawer 1                                                                                                                                                                                                                        |             | Action: open drawer 1                                                                                                                                                                                                   |
| [Reflection]                         | <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r>                                                                                                                                                |             | Observation: You open the drawer 1. In it, you see a fork 1.                                                                                                                                                            |
| Action:                              | close drawer 1                                                                                                                                                                                                                                         |             | [Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> |
| Observation:                         | You close the drawer 1.                                                                                                                                                                                                                                |             | Action: put saltshaker 1 in/on drawer 1                                                                                                                                                                                 |
| Action:                              | put saltshaker 1 in/on drawer 1                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                         |
| Observation:                         | Nothing happens.                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                         |

Example A (put a hot mug in cabinet): KnowSelf (Llama-8B) uses reflection to heat the mug; OpenAI-O1 diverges by taking an egg.  
Example B (put a saltshaker in drawer): KnowSelf uses a knowledge trigger to place the object; DeepSeek-R1 gets stuck closing the drawer first.

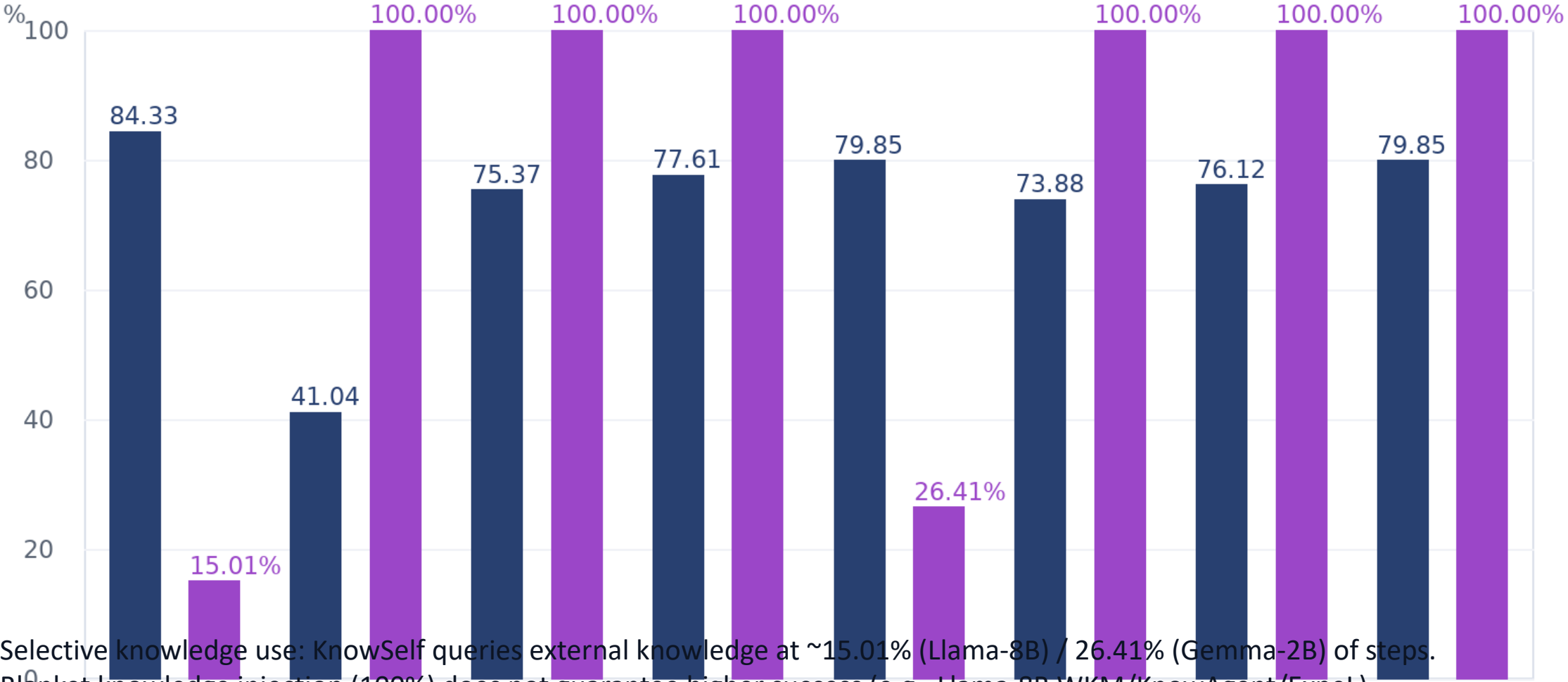
# KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using On

| Backbone | Method    | Know%  | Put    | Clean | Heat  | Cool  | Examine | Put Two | All   |
|----------|-----------|--------|--------|-------|-------|-------|---------|---------|-------|
| GPT-4o   | ReAct     | 0%     | 83.33  | 74.19 | 69.57 | 85.71 | 77.78   | 64.71   | 76.12 |
| GPT-4o   | Reflexion | 0%     | 100.00 | 87.10 | 73.91 | 90.48 | 83.33   | 70.59   | 85.07 |
| GPT-4o   | ExpeL     | 100%   | 95.83  | 83.87 | 69.57 | 80.95 | 88.89   | 52.94   | 79.85 |
| Llama-8B | ReAct     | 0%     | 33.33  | 3.23  | 0.00  | 57.14 | 66.67   | 23.53   | 27.61 |
| Llama-8B | Reflexion | 0%     | 62.96  | 22.73 | 5.88  | 64.29 | 86.36   | 50.00   | 51.49 |
| Llama-8B | ExpeL     | 100%   | 83.33  | 32.26 | 30.43 | 23.81 | 55.56   | 17.65   | 41.04 |
| Llama-8B | ETO       | 0%     | 91.67  | 70.59 | 82.61 | 61.90 | 88.89   | 64.71   | 78.36 |
| Llama-8B | KnowAgent | 100%   | 87.50  | 93.55 | 65.22 | 66.67 | 61.11   | 64.71   | 75.37 |
| Llama-8B | WKM       | 100%   | 95.83  | 87.10 | 86.96 | 61.90 | 66.67   | 52.94   | 77.61 |
| Llama-8B | KnowSelf  | 15.01% | 91.67  | 87.10 | 91.30 | 85.71 | 77.78   | 64.71   | 84.33 |
| Gemma-2B | ReAct     | 0%     | 0.00   | 9.68  | 0.00  | 4.76  | 44.44   | 0.00    | 8.96  |
| Gemma-2B | ETO       | 0%     | 91.67  | 83.87 | 78.26 | 52.38 | 77.78   | 29.41   | 71.64 |
| Gemma-2B | KnowAgent | 100%   | 91.67  | 90.32 | 69.57 | 71.43 | 66.67   | 41.18   | 73.88 |
| Gemma-2B | WKM       | 100%   | 91.67  | 87.10 | 78.26 | 71.43 | 61.11   | 52.94   | 76.12 |
| Gemma-2B | KnowSelf  | 26.41% | 87.50  | 93.55 | 73.91 | 76.19 | 83.33   | 52.94   | 79.85 |

Highlight: KnowSelf (Llama-8B) reaches 84.33% overall using 15.01% knowledge; full-knowledge methods use 100%.

# Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Adv

## Efficiency vs. Knowledge Usage



Selective knowledge use: KnowSelf queries external knowledge at ~15.01% (Llama-8B) / 26.41% (Gemma-2B) of steps.  
Blanket knowledge injection (100%) does not guarantee higher success (e.g., Llama-8B WKM/KnowAgent/Expel).

■ All success (%) ■ KnowSelf (external knowledge usage)

# Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

Indiscriminate knowledge injection is wasteful and can be suboptimal for planning.

KnowSelf formalizes situational self-awareness: fast (direct), slow (reflect), knowledgeable (retrieve).

Lightweight supervision: heuristic labeling on self-explored trajectories + special tokens; SFT then RPO.

Empirically strong: KnowSelf (Llama-8B) 84.33% vs GPT-4o ReAct 76.12% and GPT-4o Expel 79.85% on ALFWorld.

Efficiency: only 15–26% knowledge usage vs 100% for full-knowledge methods; generalizes across model scales (Gemma-2B → Llama-8B).

Future directions (from framework motivation): extend situational self-awareness beyond knowledge retrieval to broader tool use and general interactive en